

Technology Stack

Date	17 JULY 2026
Team ID	SWTID-2026-4144
Project Name	RISING WATERS
Maximum Marks	2 Marks

Define Your Technology Stack

The Rising Waters project uses modern technologies to build a reliable flood monitoring and early warning system with secure data processing and real-time visualization.

S.No	Layer	Technology	Purpose
1	Frontend	React, HTML5, CSS3	Responsive dashboard
2	Backend	Django REST	Secure APIs
3	Database	PostgreSQL	Sensor data storage
4	Cloud	AWS	Scalable deployment
5	Version Control	GitHub	Code collaboration
6	APIs	Google Maps, OpenWeather	Maps & weather

Detailed Technology Justification

- Frontend: React provides a responsive and interactive user interface.
- Backend: Django REST develops secure APIs and application logic.
- Database: PostgreSQL stores flood and sensor information efficiently.
- Cloud: AWS ensures reliable hosting and scalability.
- Version Control: GitHub supports collaboration and version tracking.
- Third-Party APIs: Google Maps and OpenWeather APIs provide mapping and weather information.